



UKUDLA Data Science Certificate

Data Integration

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UKUDLA
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Systems and Applied Agricultural
and Food Data Science

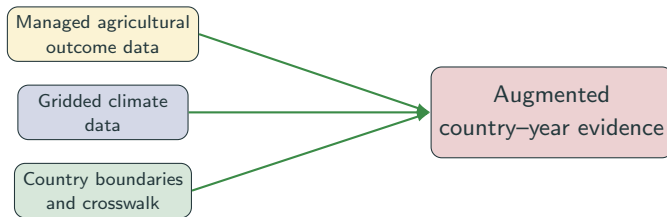


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A managed dataset may not answer the question



One managed source describes agricultural outcomes, but not the environmental conditions associated with them. A second source may add useful evidence—if the sources can be aligned responsibly.

Data integration combines information from different sources into a coherent unit of analysis while preserving meaning and lineage.

Data integration extends data management

Manage each source

Describe, validate and preserve both the existing and additional data.

Integrate deliberately

Align identifiers, grains, periods, spaces, units and classifications.

Manage the result

Audit the join and document the new artifact, assumptions and lineage.

This session moves from **motivation**, through the essential **concepts**, to an auditable **application** across compatible sources.

Data acquisition supports this process; integration is the teaching focus.

Sources bring different representations

Agricultural statistics

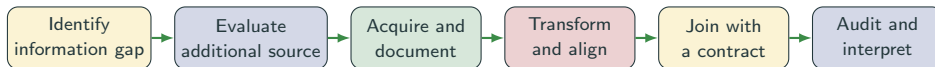
- tabular country–year observations;
- yield, production and harvested area;
- provider labels, units and flags; and
- a fixed teaching snapshot.

Gridded precipitation

- daily gridded precipitation estimates;
- country polygons define spatial support;
- October–April defines a season; and
- API responses require aggregation.

Different representations must be made compatible without hiding how their meaning, space and time differ.

A data need initiates integration



The research question determines what information is missing, which source is appropriate, and what the integrated observation should represent.

Do not acquire an additional dataset before defining why and how it will be integrated.

Compatibility has several dimensions

Structural compatibility

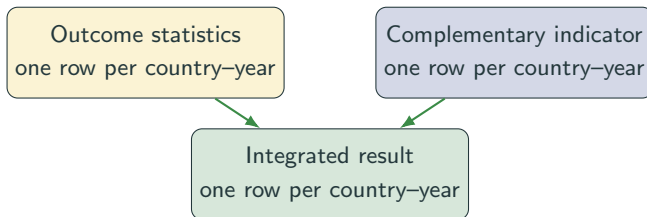
- identifiers and candidate keys;
- dataset grain and relationships;
- variable types and units;
- missing-value conventions; and
- classifications and code lists.

Scientific compatibility

- population and geographic support;
- observation and reporting periods;
- measurement definitions;
- aggregation assumptions; and
- fitness for the intended question.

A technically successful join does not prove that the combined variables refer to comparable evidence.

State the grain before joining



- A one-to-one join should preserve the expected key and row count.
- A one-to-many relationship can legitimately multiply rows.
- An unexpected many-to-many relationship often indicates an incomplete key.
- The join function cannot decide whether the scientific relationship is valid.

Stable identifiers and crosswalks connect sources

Names are fragile

- spelling and punctuation differ;
- abbreviations differ;
- names and boundaries change; and
- the same name can identify different entities.

Provider label	Stable ID
South Africa	ZAF
Zimbabwe	ZWE
Zambia	ZMB

A crosswalk is a documented dataset, not an invisible correction embedded in code.

Check unmatched keys, duplicated keys, new missing values and row multiplication after every join.

Integration requires alignment, not only a join

Dimension	Possible mismatch	Decision to document
Identifiers	FAO, ISO and provider-specific codes	Crosswalk and valid period
Time	Day, month, calendar year, growing season	Aggregation and period definition
Space	Points, districts, countries, raster cells	Boundary, CRS, extent and summary rule
Units	Tonnes, hectares, kg/ha, indices	Conversion and dimensional consistency
Schema	Names, types, categories and flags	Mapping and retained information

Converting daily gridded values into regional growing-season totals is an analytical choice, not a neutral format conversion.

Validate the result and preserve lineage

Join audit

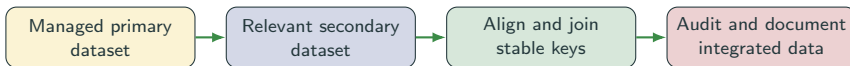
- expected and actual row counts;
- matched and unmatched keys;
- duplicated or multiplied records;
- missingness introduced; and
- resulting grain and coverage.

Lineage record

- source of every retained variable;
- crosswalks and conversion rules;
- temporal or spatial aggregation;
- script and input versions; and
- known losses and limitations.

A critical failed expectation should stop the pipeline visibly rather than produce a plausible-looking result silently.

Key message: integration requires explicit alignment



1. Identify the additional information required by the question.
2. Reconcile grain, identifiers, time, space, definitions and units.
3. Map source labels to stable analytical identifiers.
4. Join on tested keys under an explicit integration contract.
5. Save the derived table, audit, dictionary and lineage record.

A successful join is not sufficient: integrated data inherit every source and alignment limitation.

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